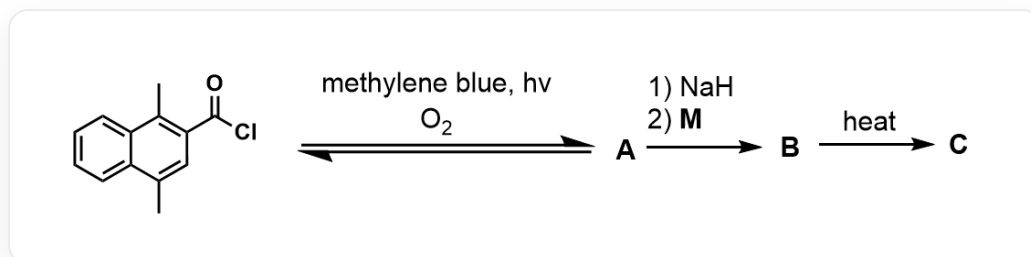


Question

The following reactions are known:

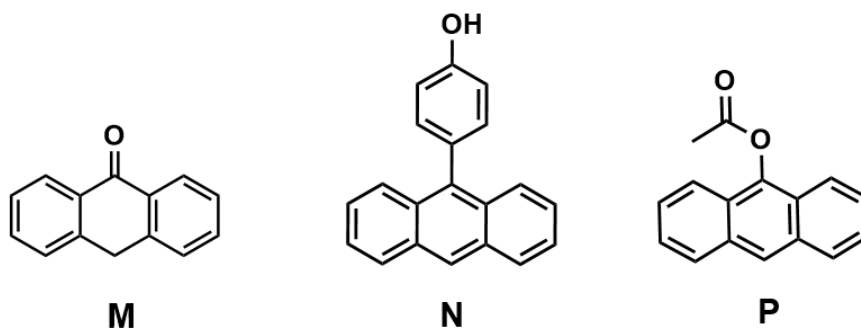


CC1=C(C(Cl)=O)C=C(C)C2=CC=CC=C21 undergoes a reversible reaction with O_2 under the conditions of methylene blue and light, yielding compound **A**. Subsequently, **A** first reacts with NaH, then with **M** to form **B**. Finally, **B** generates **C** under heating conditions.

Here, **B** and **C** are isomers. Regarding this reaction, the following experimental facts are observed:

(1) When **M** is replaced with **N** while keeping other conditions unchanged, the corresponding compound analogous to **B** can still be obtained with a high yield. However, in the final step of the reaction, a large amount of oxygen is released, and almost no compound corresponding to **C** is produced.

(2) During the transformation from **B** to **C**, if **P** is added, the oxidation product **Q** of **P** can be detected, and the yield of **C** is almost the same as when **P** is not added.



M is O=C(C1=CC=CC=C1C2)C3=C2C=CC=C3; **N** is OC(C=C1)=CC=C1C2=C3C(C=CC=C3)=CC4=CC=CC=C42; **P** is CC(OC1=C2C=CC=CC2=CC3=C1C=CC=C3)=O

Which of the following statements is incorrect? (If all are correct, select F.)

- A.** The number of chiral carbons in **A** is 2.
- B.** There are **H** atoms in **B** existing in 12 different chemical environments.
- C.** There are 11 types of hydrogen chemical environments in **C**.
- D.** **B** → **C** is dominated by intramolecular reactions
- E.** After replacing **M** with **N**, no corresponding **C** is generated because the spatial distance is too far to achieve oxygen transfer.
- F.** All of the above options are correct

Answer

Correct Answer: **C**

Detailed Explanation

First, the reaction process is inferred. Since the question states, "but a large amount of oxygen is released in the final step of the reaction," the first step must involve an addition reaction with O_2 , forming $CC12C=C(C(C3=C1C=CC=C3)(OO2)C)C(Cl)=O$, with 2 chiral carbons, making option A correct.

CHECKPOINT

1 PTS

The structure of **A** is $CC12C=C(C(C3=C1C=CC=C3)(OO2)C)C(Cl)=O$

Since **M** is essentially anthranol, NaH will react with the phenolic hydroxyl group, turning **M** into a nucleophile. Subsequently, it reacts with **A** to form $CC12C=C(C(C3=C1C=CC=C3)(OO2)C)C(OC4=C5C=CC=CC5=CC6=C4C=CC=C6)=O$, meaning **B** has 12 chemically distinct hydrogens, making option B correct.

CHECKPOINT

1 PTS

The structure of **B** is $CC12C=C(C(C3=C1C=CC=C3)(OO2)C)C(OC4=C5C=CC=CC5=CC6=C4C=CC=C6)=O$

Since the question hints that "**B** and **C** are isomers," and $B \rightarrow C$ is a heating process, the transformation yields the thermodynamic product. Structures with more complete benzene rings are more stable, so $B \rightarrow C$ converts two complete benzene rings into three, meaning **C** is

CC1=CC(=C(C)C2=C1C=CC=C2)C(=O)OC34C5=C(C=CC=C5)C(C6=C3C=CC=C6)OO4, and **C** has 12 chemically distinct hydrogens, making option C incorrect.

CHECKPOINT

1 PTS

The structure of **C** is
CC1=CC(=C(C)C2=C1C=CC=C2)C(=O)OC34C5=C(C=CC=C5)C(C6=C3C=CC=C6)OO4

According to the question, "when **M** is replaced with **N**, under otherwise identical conditions, the compound corresponding to **B** is obtained in high yield, but a large amount of oxygen is released in the final step of the reaction, and almost no compound corresponding to **C** is produced." The key difference between **M** and **N** lies in the fact that, in the corresponding product, the anthracene moiety in the **N**-derived structure is farther from the peroxide bond. Thus, the inability to rearrange and the release of oxygen instead indicate that **B** → **C** is an intramolecular reaction and is sterically controlled, making options D and E correct.

CHECKPOINT

1 PTS

B → **C** is an intramolecular reaction

In conclusion, the answer is C.